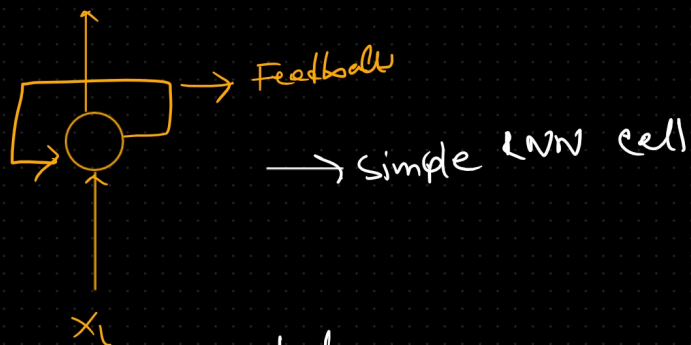


## \* Agenda:

- ① LSTM → with practical
- ② GRU →
- ③ BiRNN, BiLSTM, BiGRU
- ④ Hate speech classification using LSTM

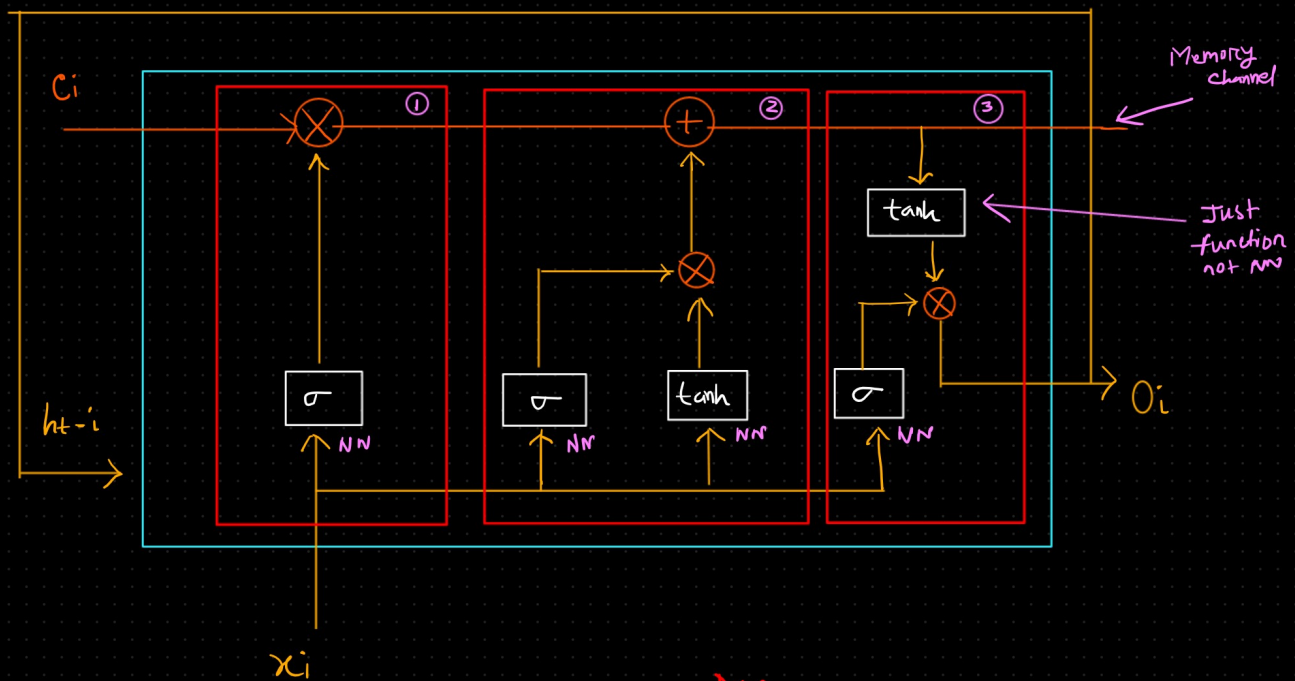
## # RNN:



- ① Can't remember longer context

LSTM → Long short term memory

# LSTM Architecture



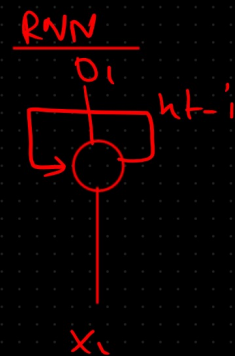
Here,

$x_i$  = Input

$O_i$  = Output

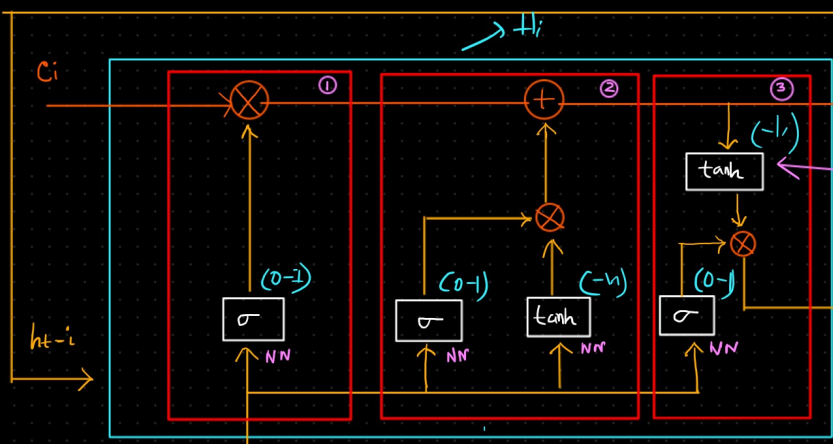
$c_i$  = memory channel

$h_{t-i}$  = Input from previous output



(4)

10-0001  $x_i$



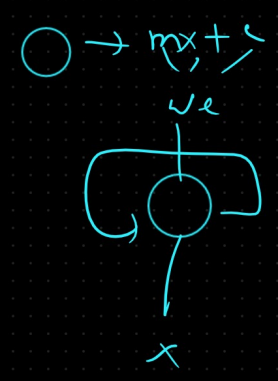
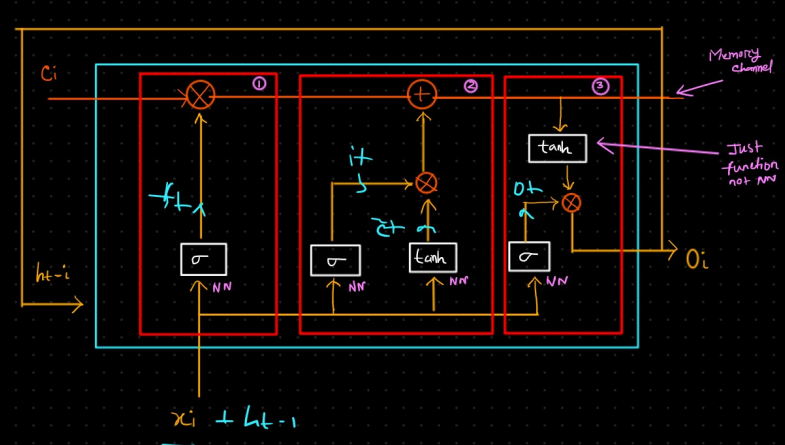
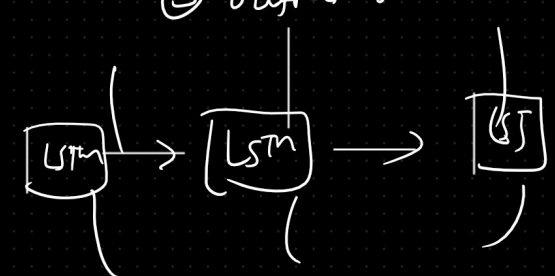
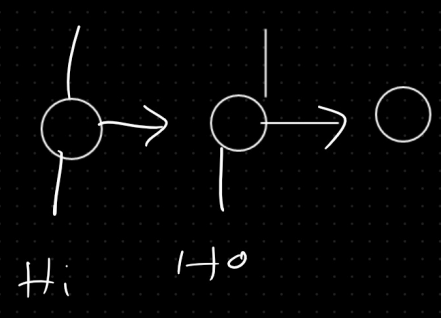
Memory channel  
Just function not NN

$\begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix}$   
 $\begin{bmatrix} 0.9 \\ 0.3 \end{bmatrix}$  → mem  
 $\begin{bmatrix} 0.9 \\ 0.3 \end{bmatrix}$  → res  
 $H_i, H_{i-1}, H_{i-2}, \dots$   
 $H_i \cdot H_{i-1} \cdot 0.6$   
 $C_i = \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$   
 $\oplus H_i, H_{i-1} 0.3 \times 0.001$   
 $\alpha_i = 0.3 \Rightarrow 0$   
 $+ 0.6$   
 $\otimes \rightarrow \text{Multiply}$   
 $\oplus \rightarrow \text{Addition}$

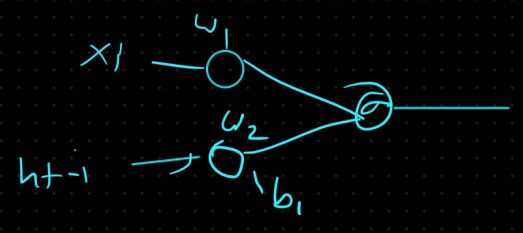
How are you

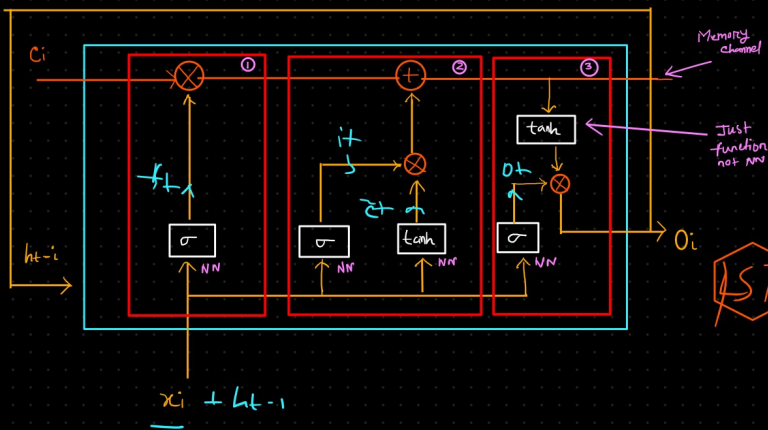
$\begin{bmatrix} 0.6 \\ 0.6 \end{bmatrix} \times 1 = 0.6$   
 $x_i = H_i \Rightarrow 0.6$   
 $0.6 \times -1 = -0.6$   
components

- ① forget gate
- ② Input gate or memory gate
- ③ output gate



$$f_t = \sigma(w_f[h_{t-1}, x_t] + b_f)$$





LST

sentiment anal → GPO

chat completion

Hi how are you? LSTM

$$i_t \Rightarrow \sigma(w_t[h_{t-1}, x_t] + b_t)$$

$$z_t \Rightarrow \tanh(w_t[h_{t-1}, x_t] + b_t)$$

$$o_t \Rightarrow \sigma(w_t[h_{t-1}, x_t] + b_t)$$

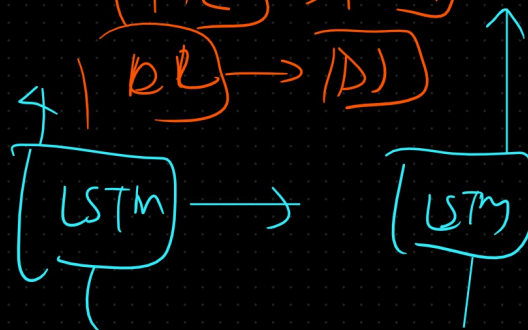
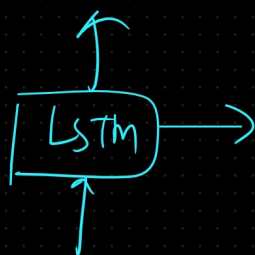
$$c_t \Rightarrow o_t \times \tanh(c_t)$$

⇒ final

Ant → les  
shot gun

Tiger → les  
P.H  
short

ML → good → les  
BL → DJ



is

good

S5nol This

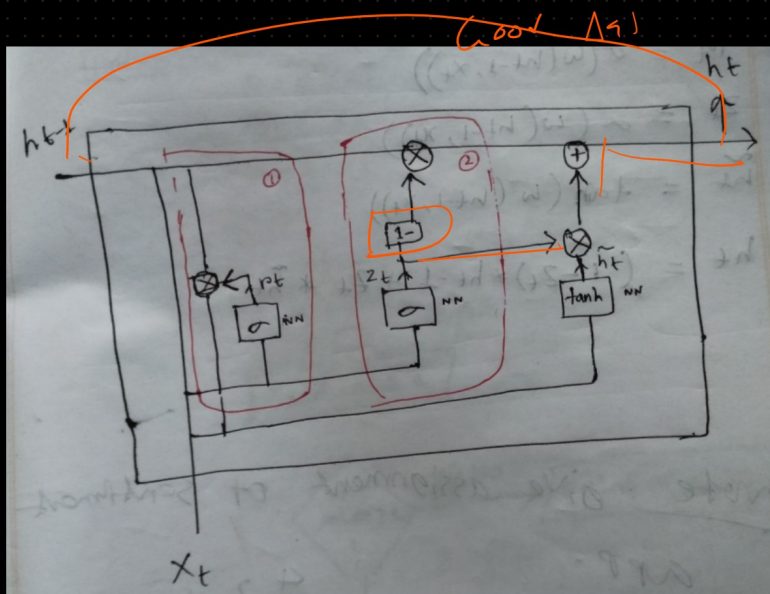
N < 0.5 > P

softwars

0 → 0.99  
0 → 0.88  
0 → 0.77

# CPU

## Gated Recurrent Unit



1 -

Inp 1 -

$x_t \rightarrow$  Input  
 $h_{t-1} \rightarrow$  prev output as input

- ① forget gate
- ② update gate

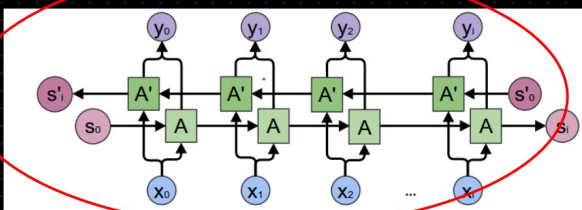
RNN



How are you?



LSTM, GRU



B: RNN

